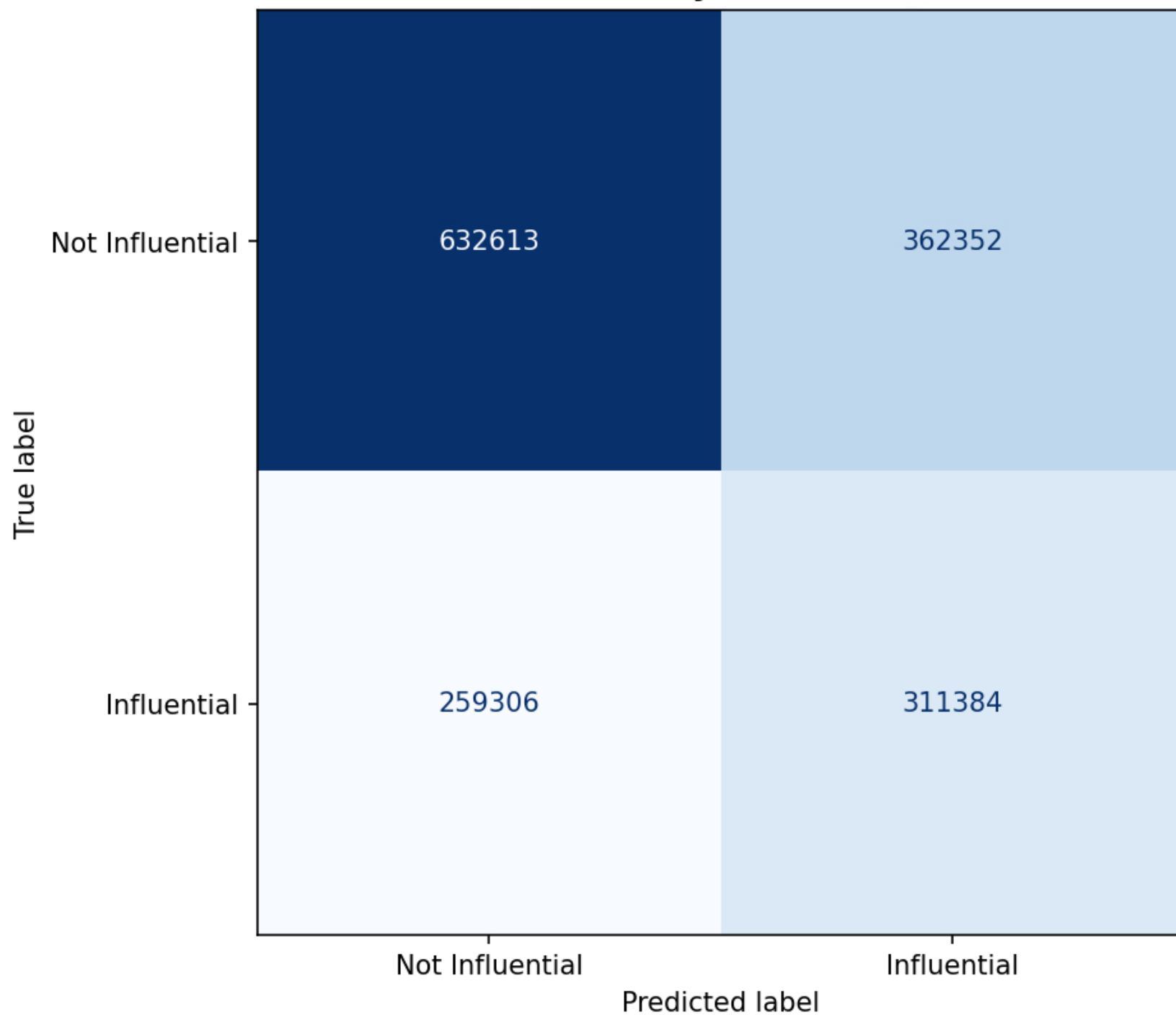
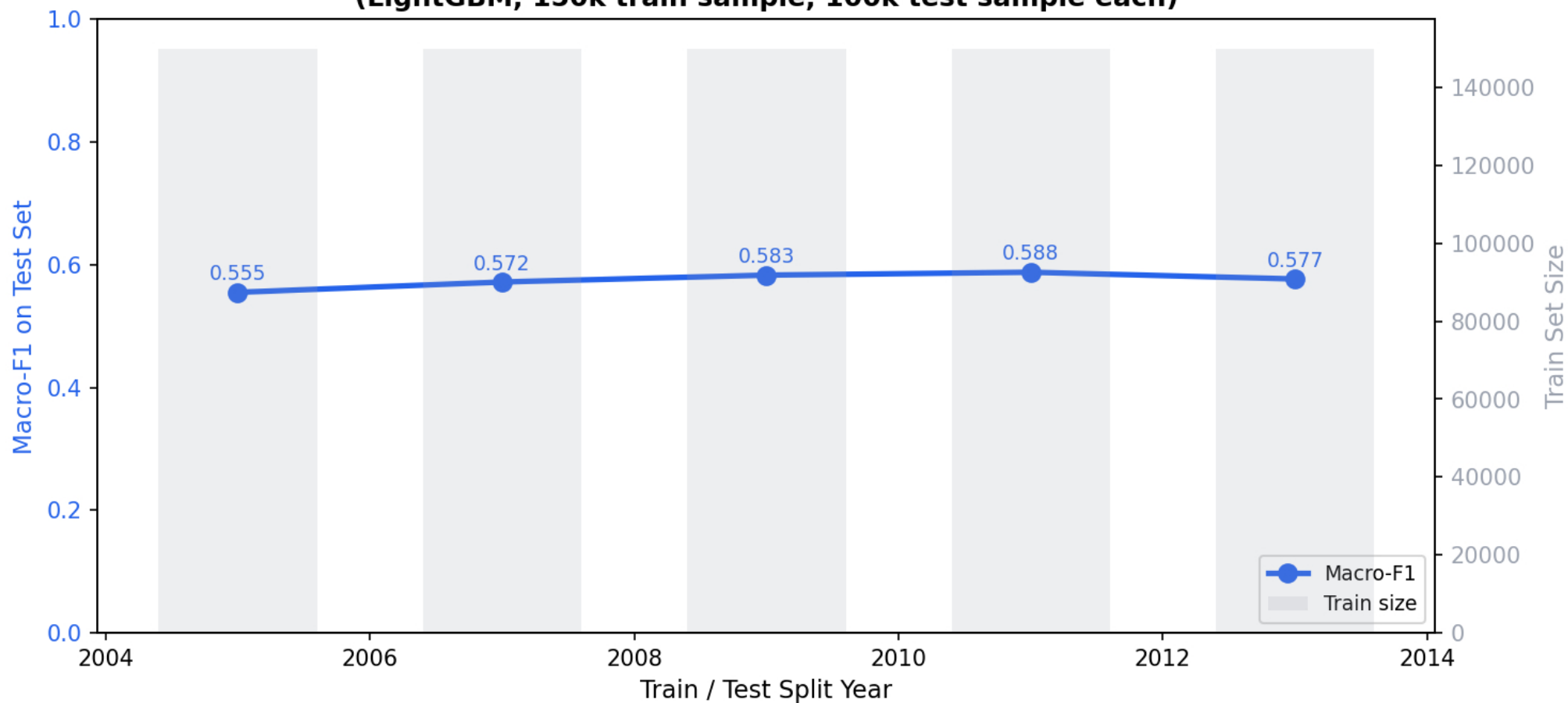


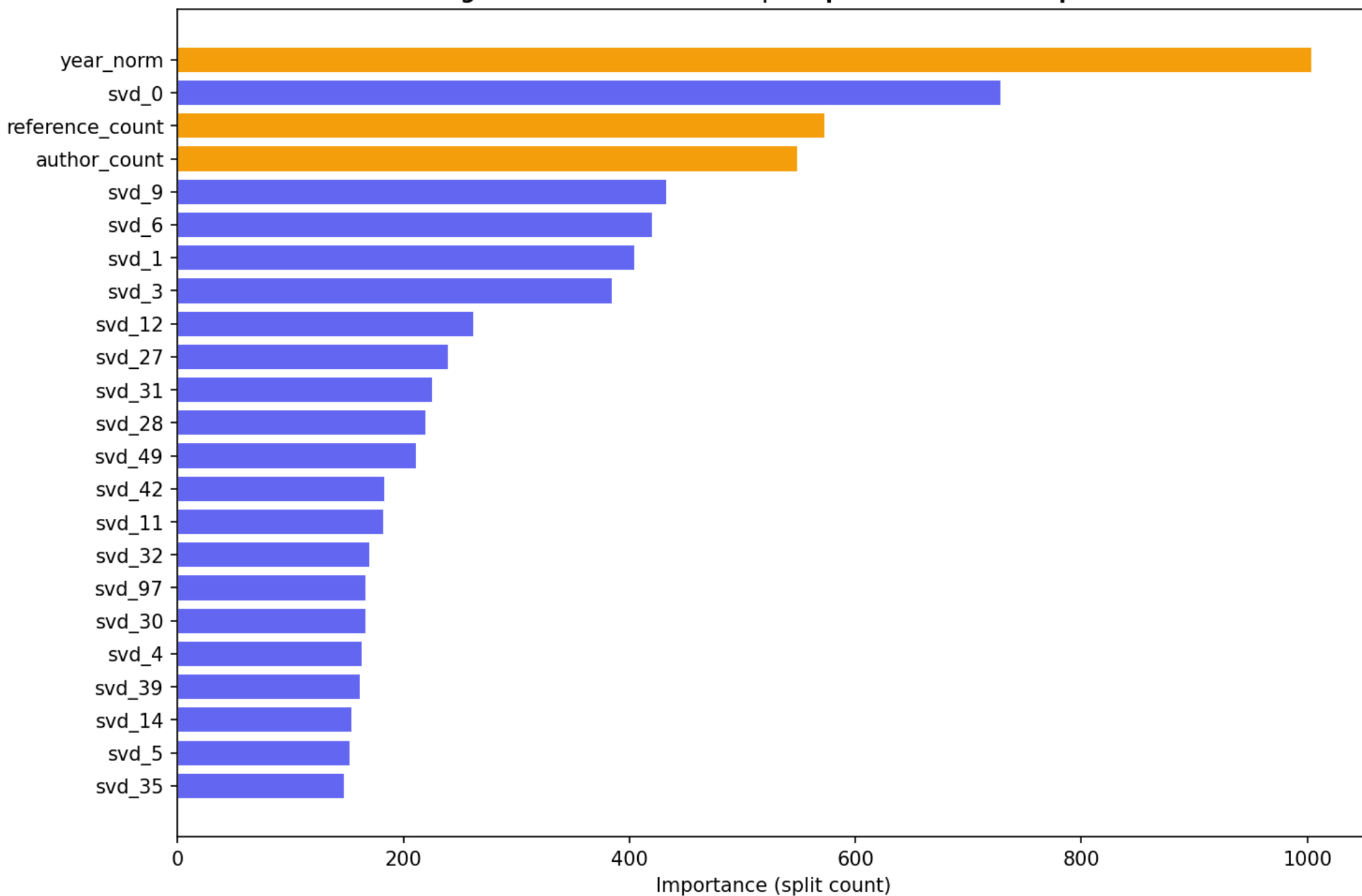
Confusion Matrix — LightGBM
(test set: year $\geq$ 2010)



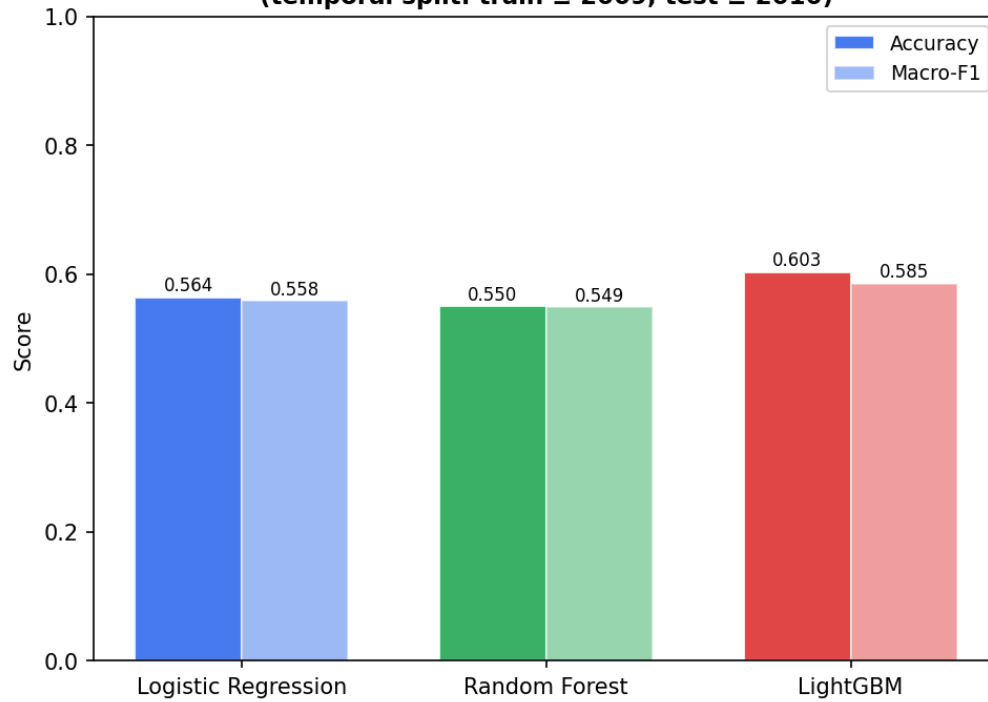
Model Performance vs. Train/Test Year Threshold
(LightGBM, 150k train sample, 100k test sample each)



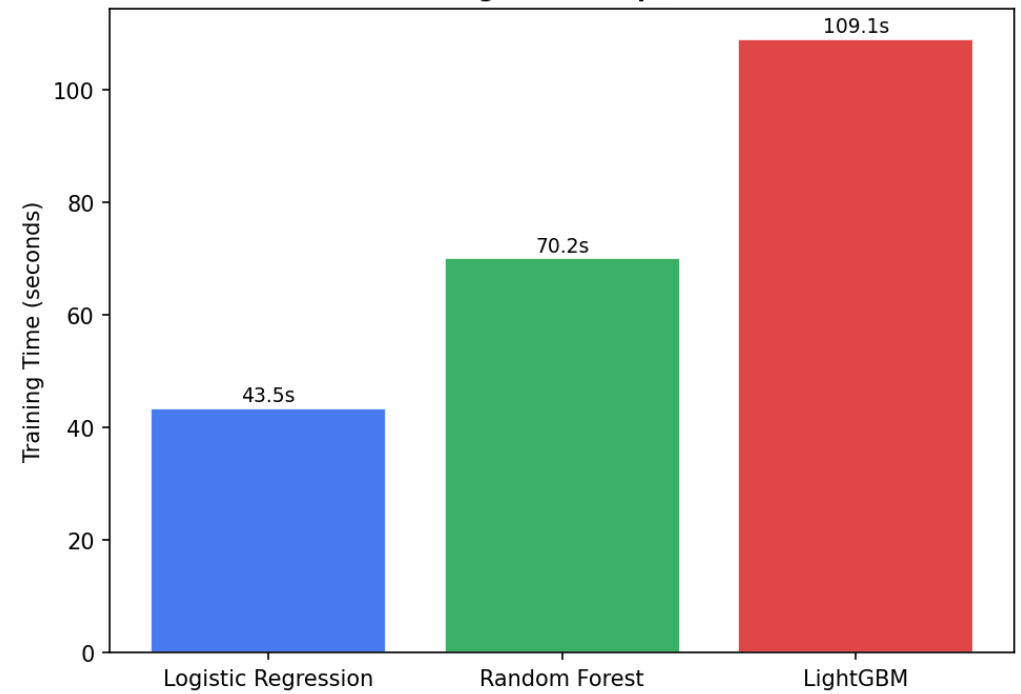
LightGBM Feature Importance (top 23)
Orange = structured feature | Purple = SVD text component



Model Accuracy & Macro-F1
(temporal split: train $\leq$ 2009, test $\geq$ 2010)

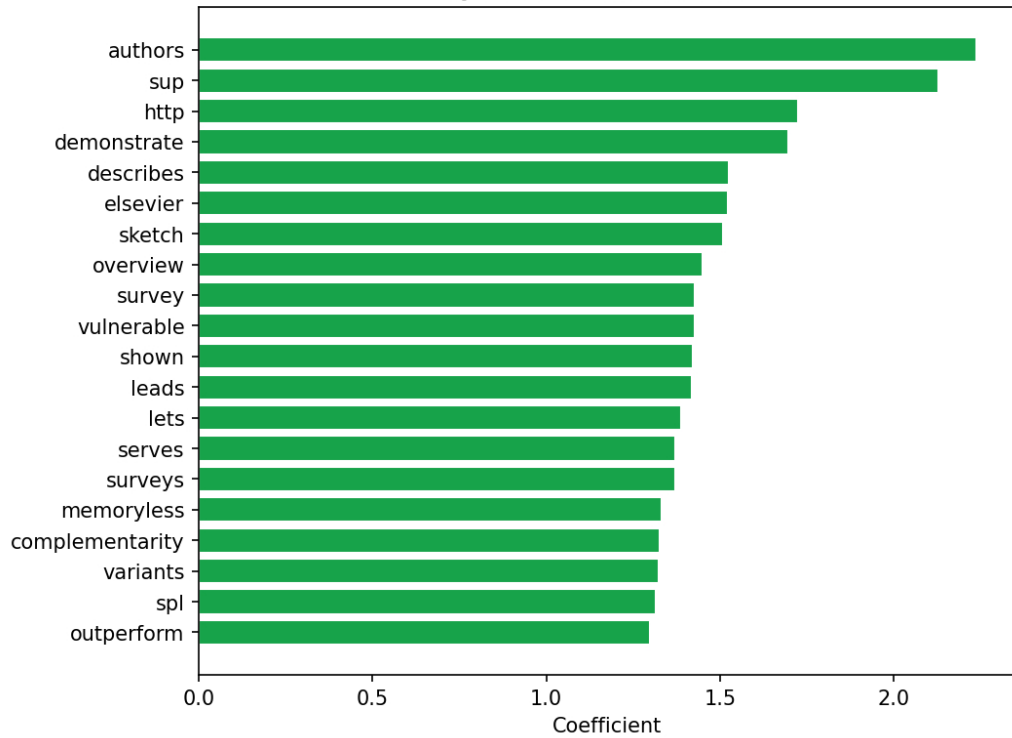


Training Time Comparison

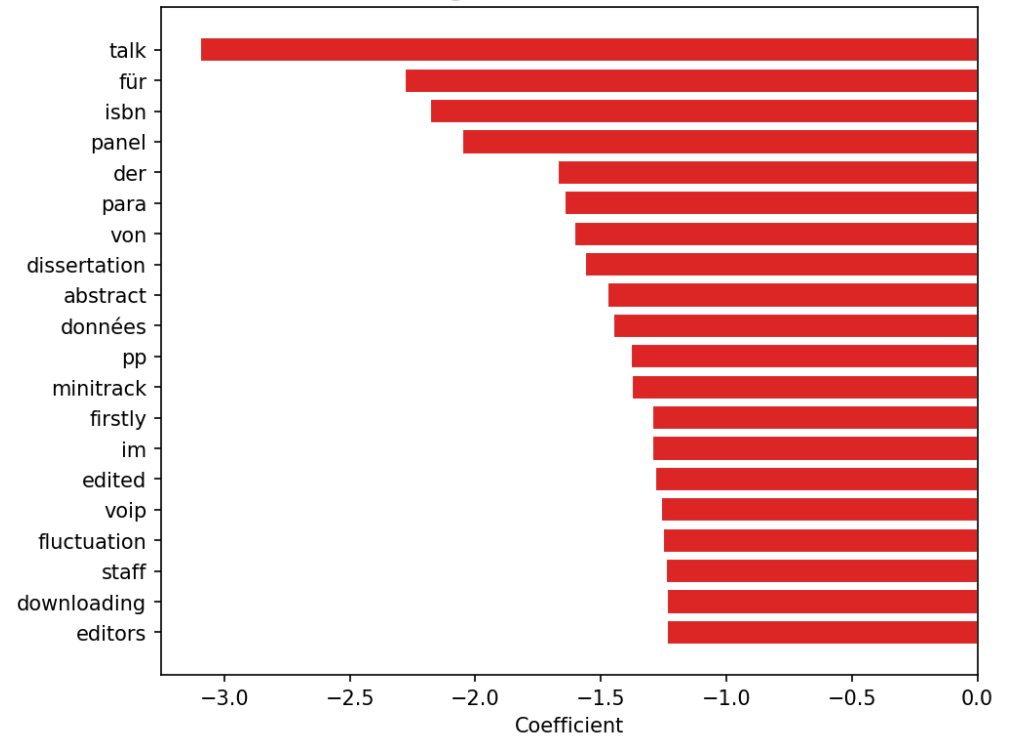


Logistic Regression — Most Predictive Keywords

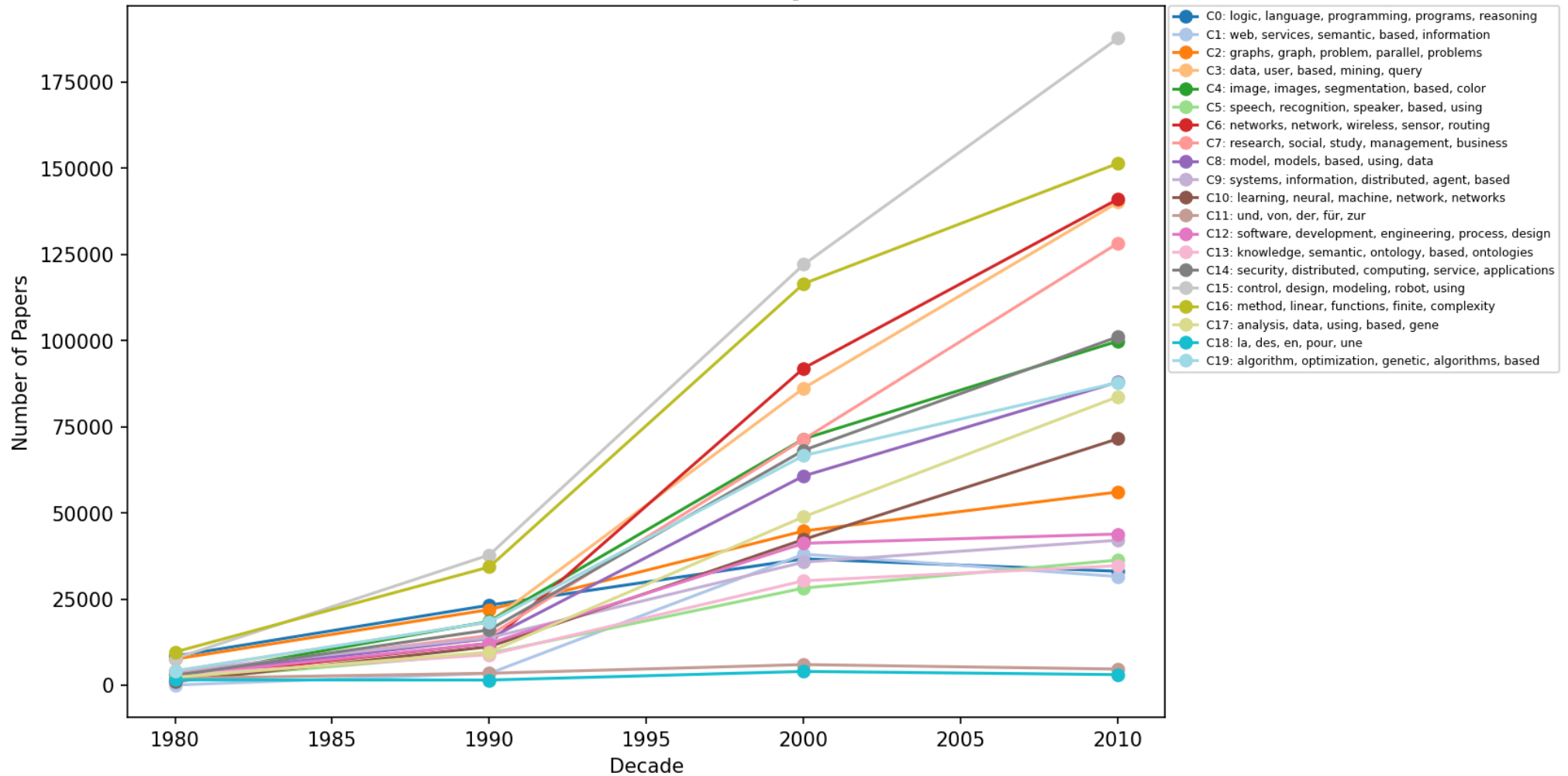
Top 20 Keywords → Influential
(positive LR coefficients)



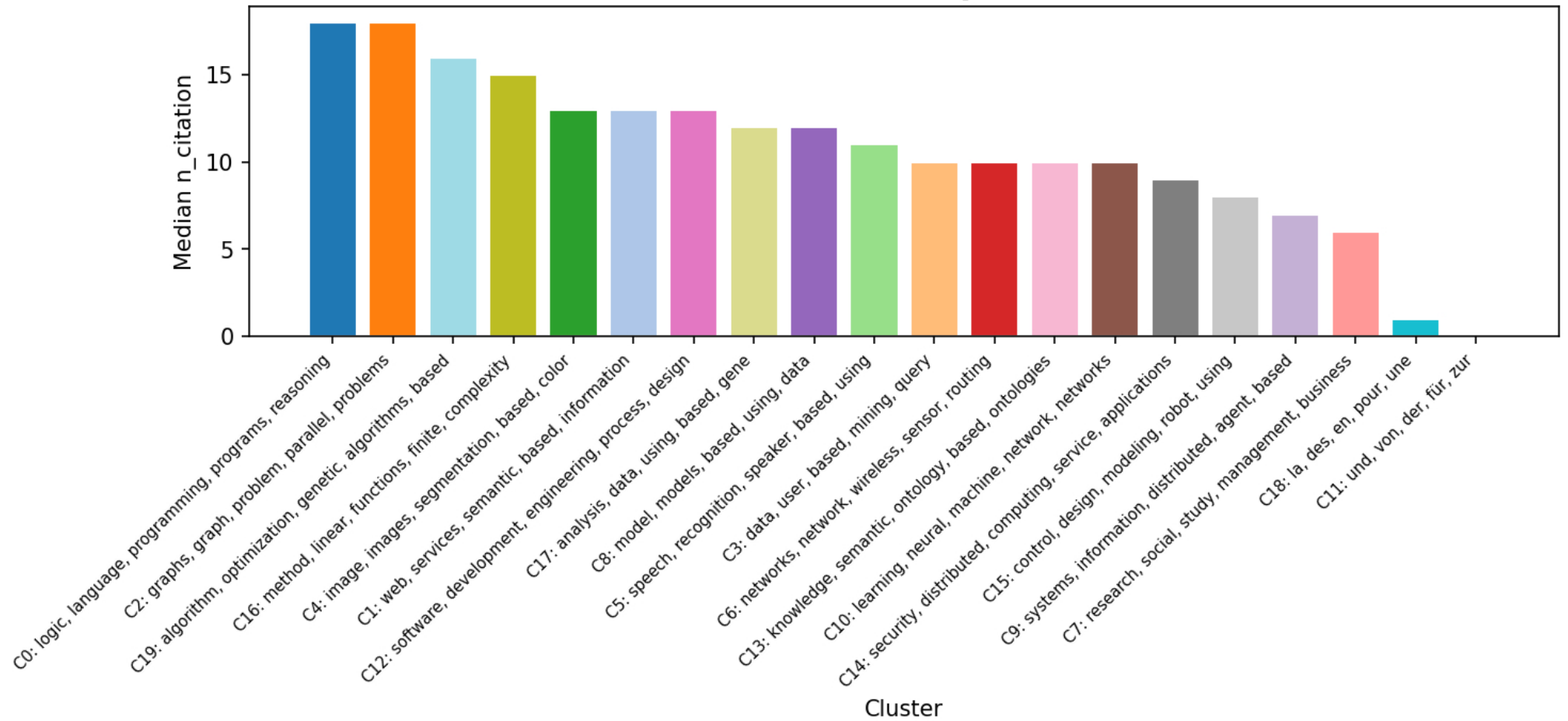
Top 20 Keywords → Not Influential
(negative LR coefficients)



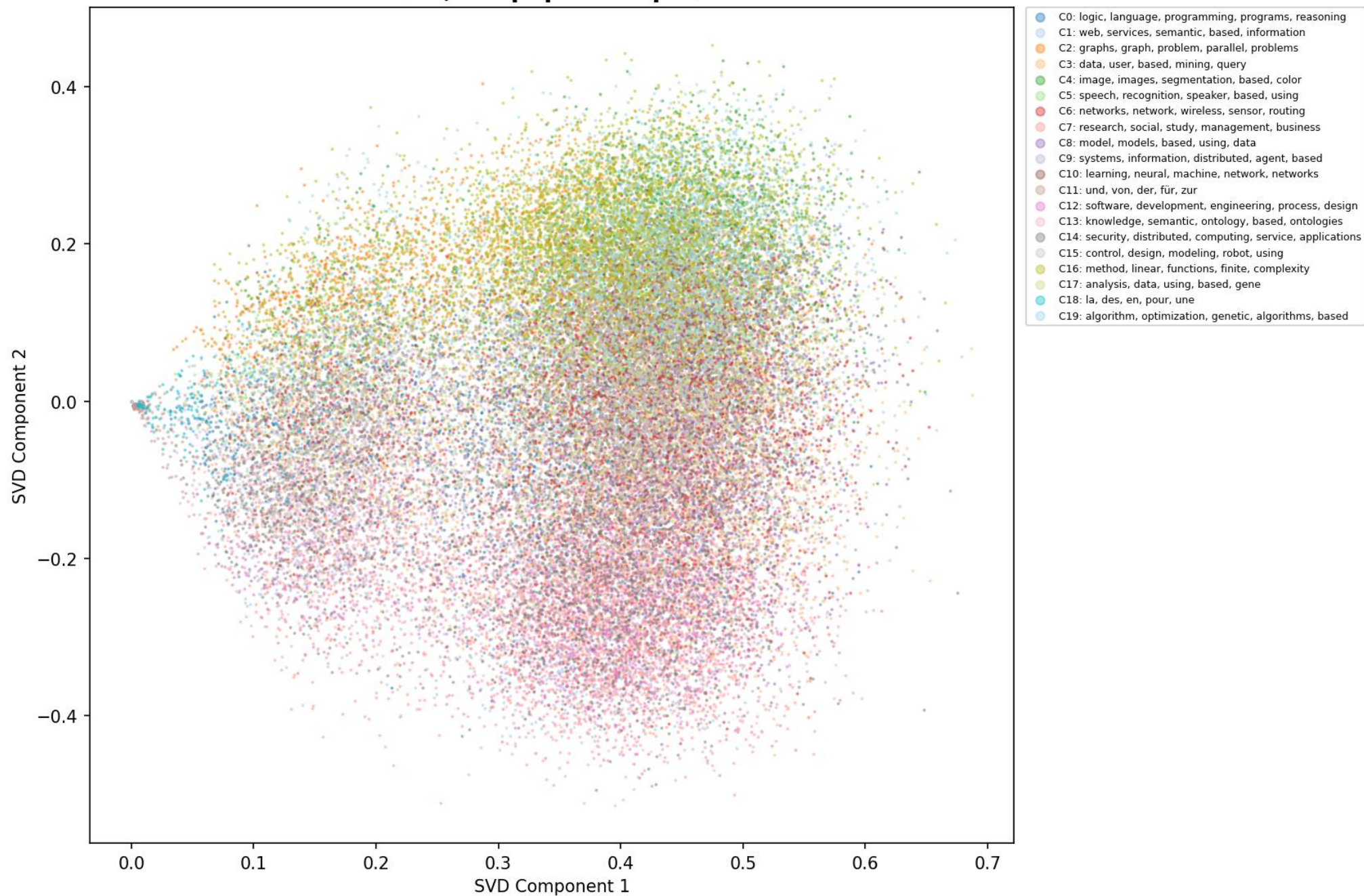
Cluster Growth Over Time (by Decade)



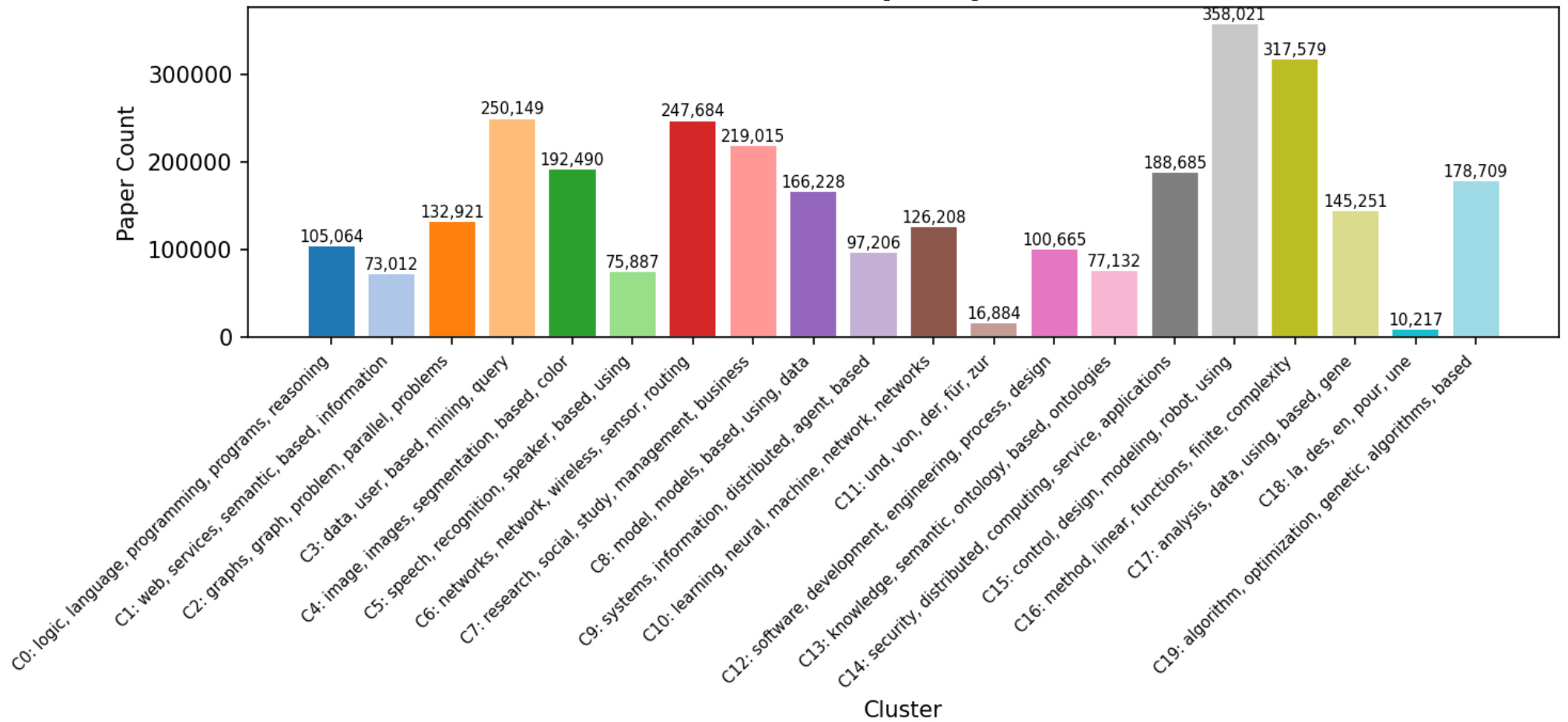
Median Citations per Cluster



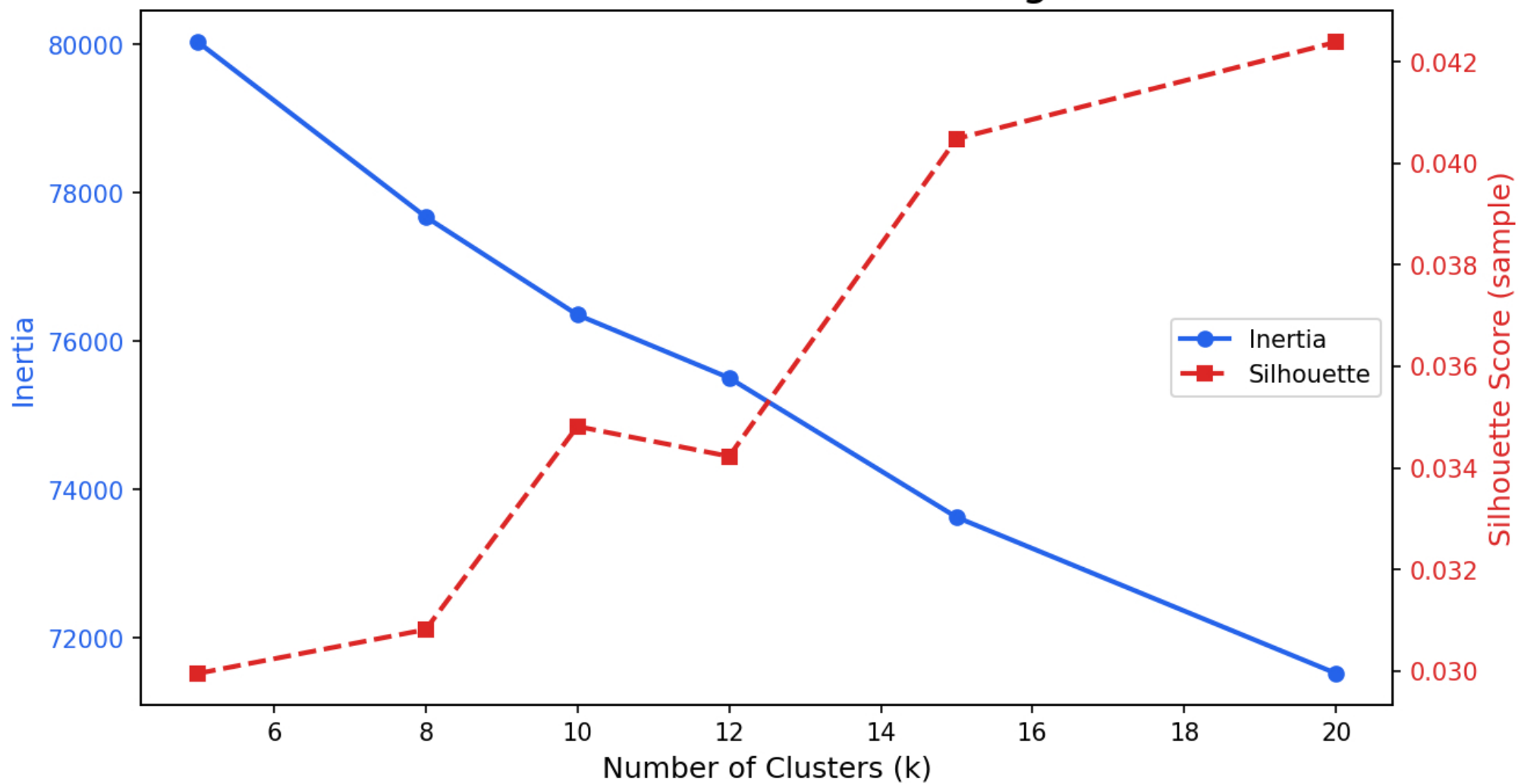
**Paper Clusters (k=20) — SVD 2D Projection
(50,000 paper sample)**



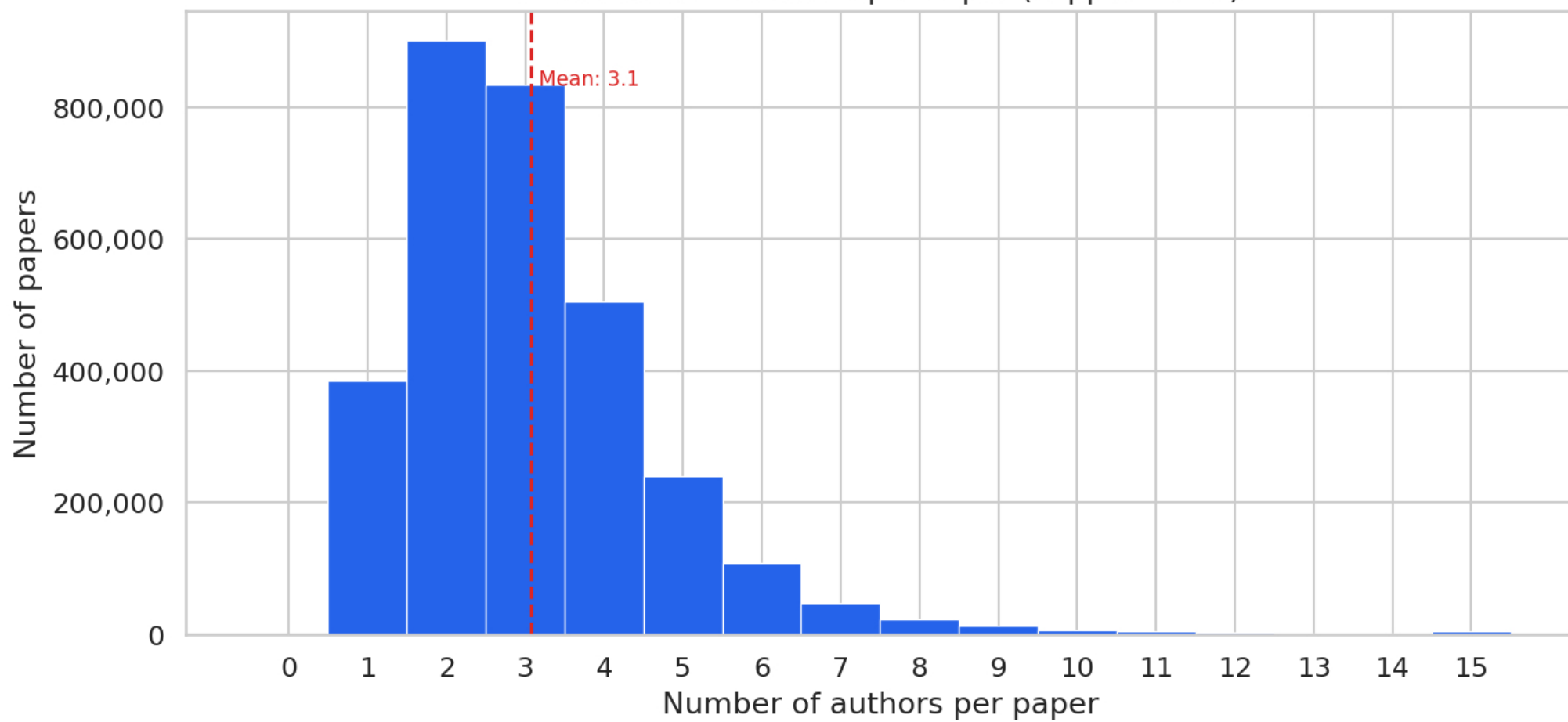
Number of Papers per Cluster



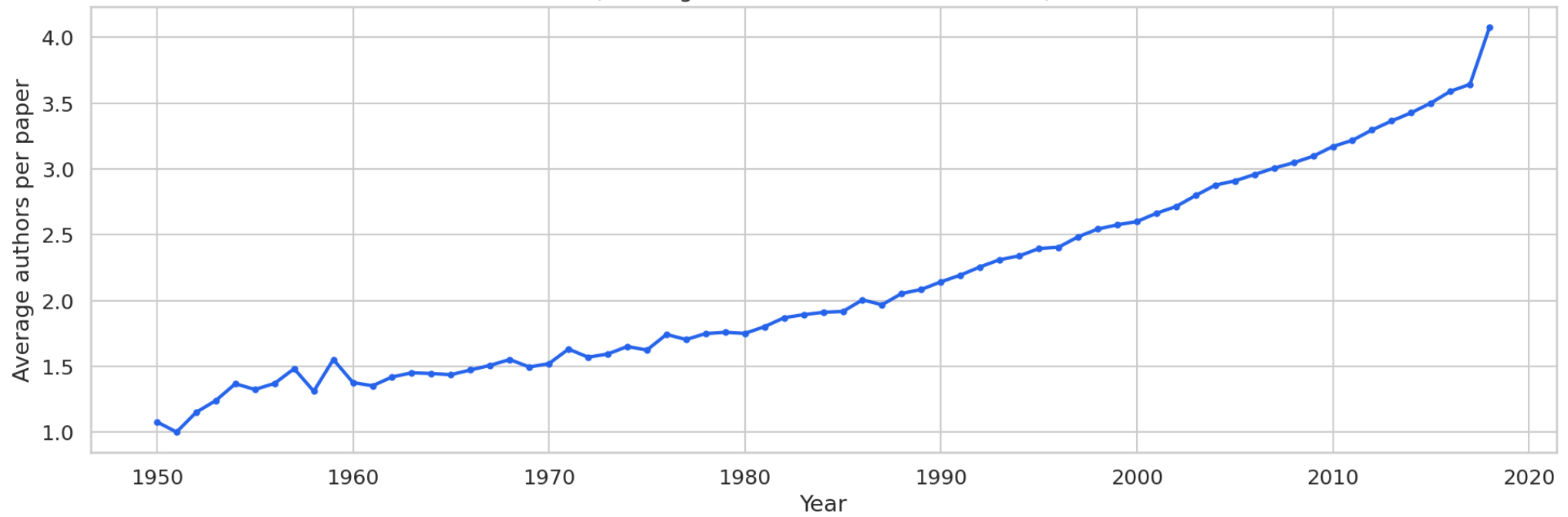
Elbow & Silhouette — Choosing k



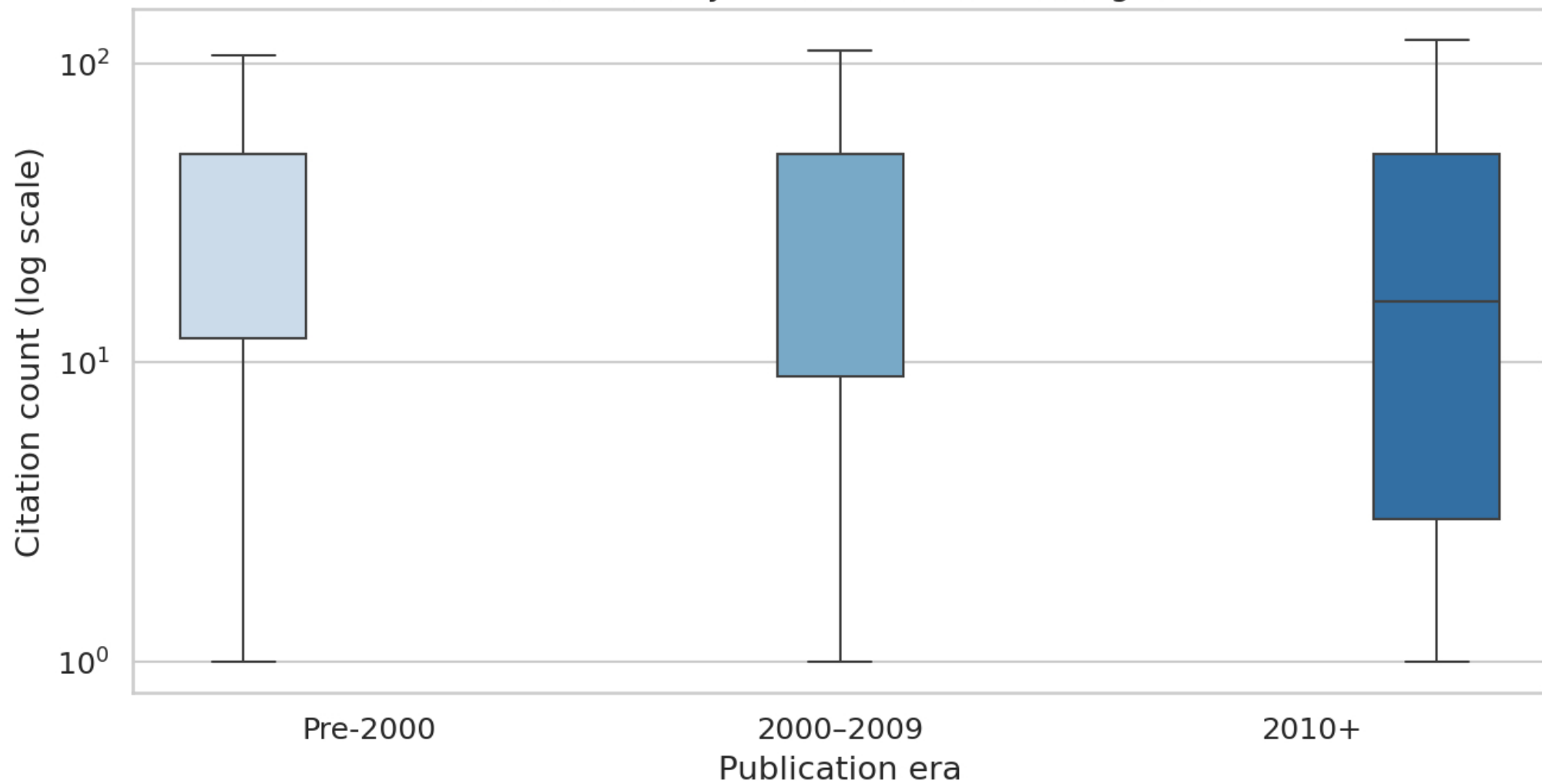
Distribution of Authors per Paper (capped at 15)

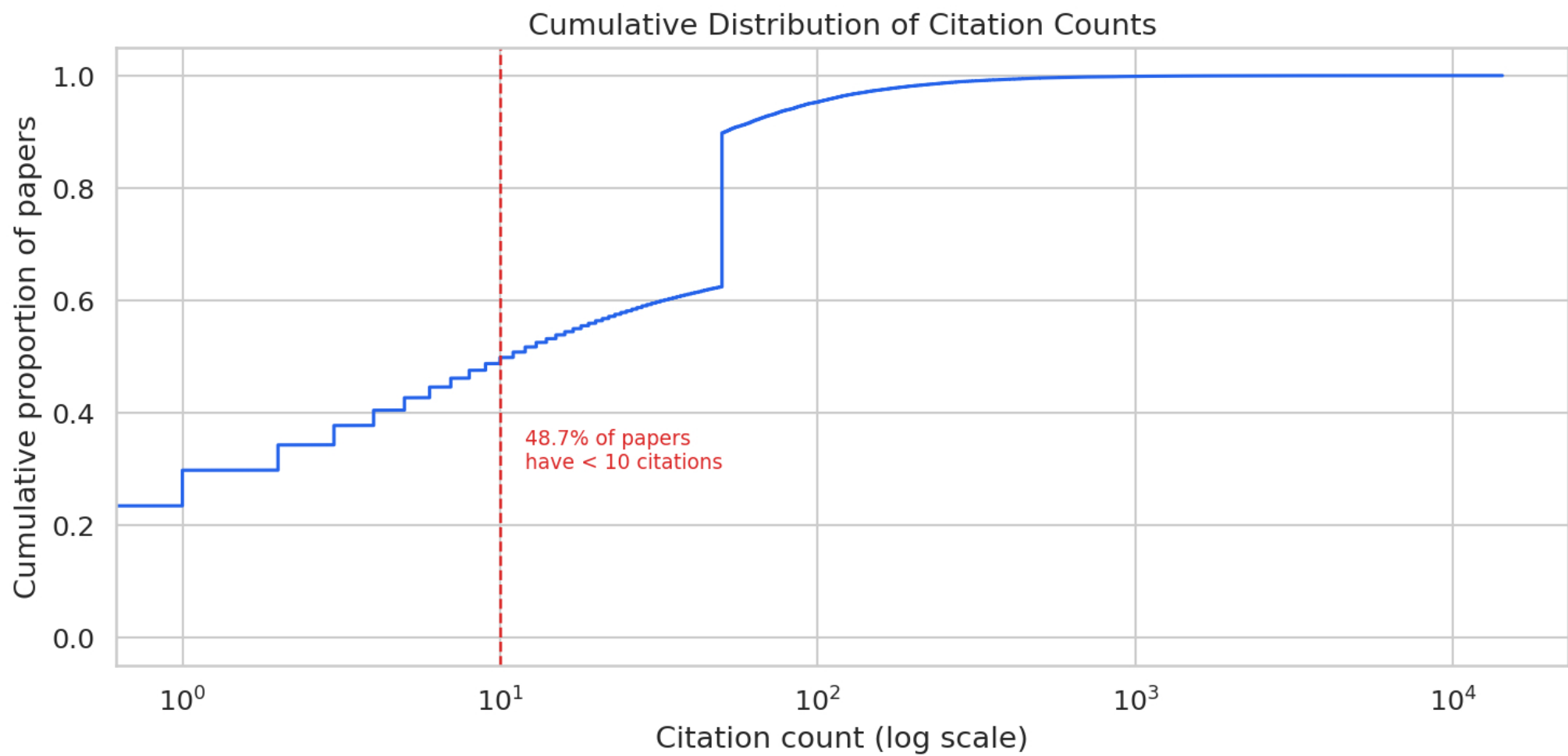


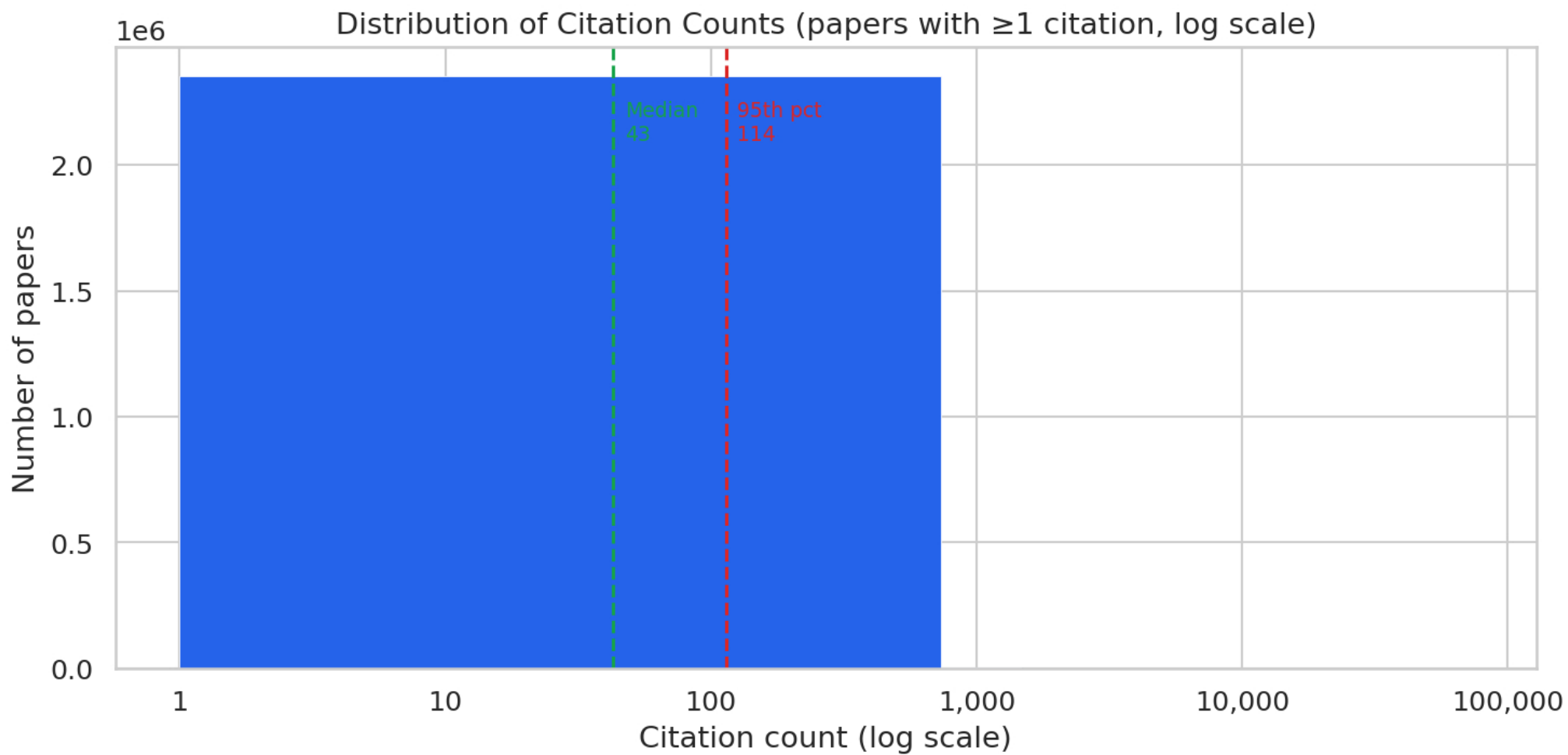
Average Number of Authors per Paper Over Time
(tracks growth of collaborative research)



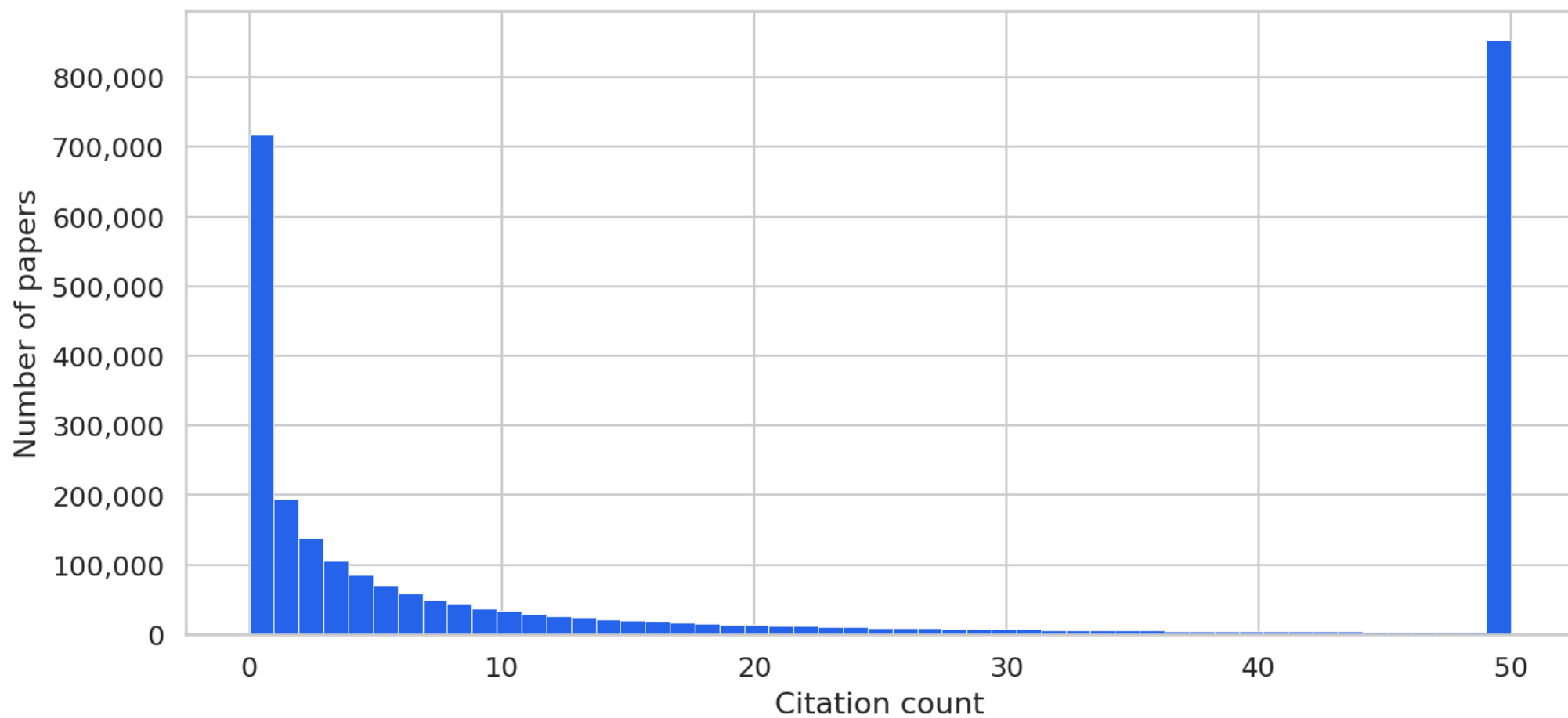
Citation Count by Era (outliers hidden, log scale)



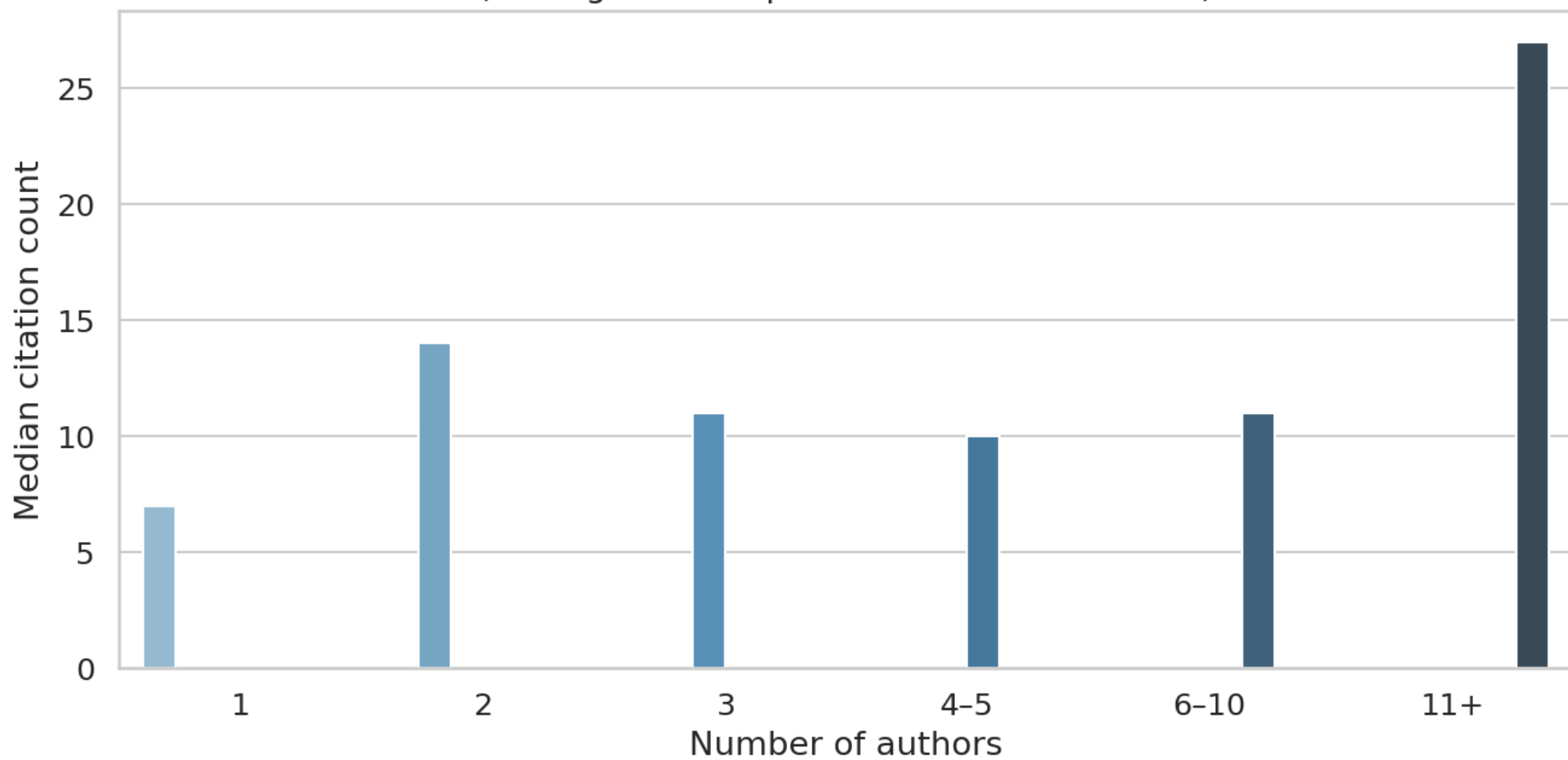




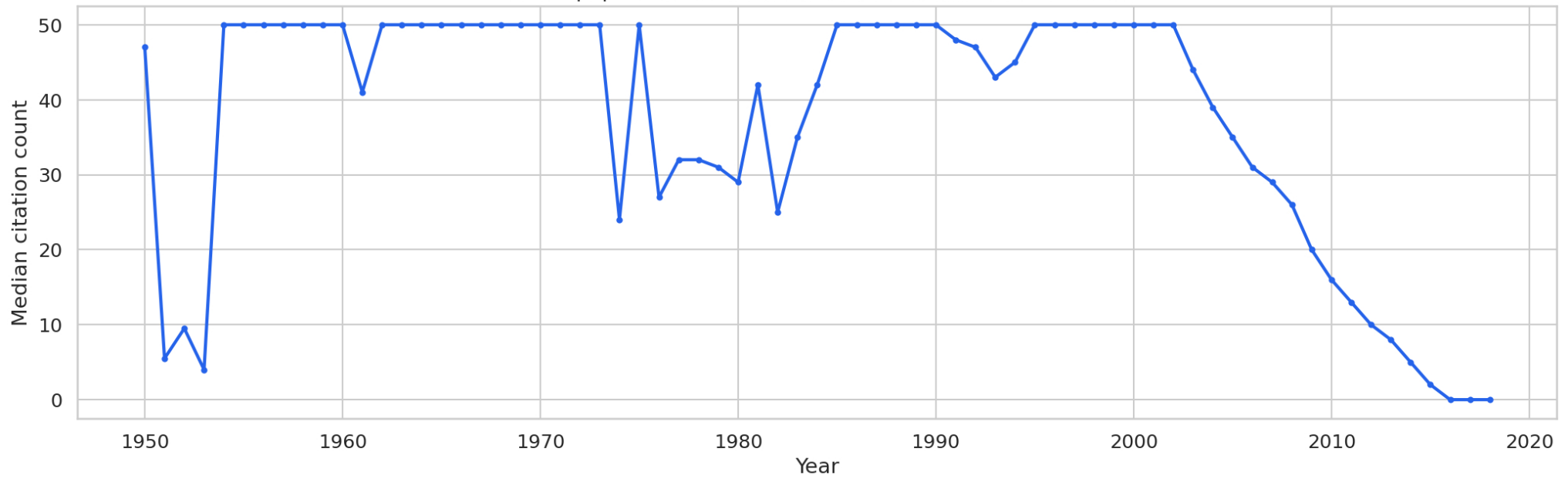
Citation Count Distribution — Zoomed: 0 to 50 Citations



Median Citations by Author Count
(do larger teams produce more-cited work?)

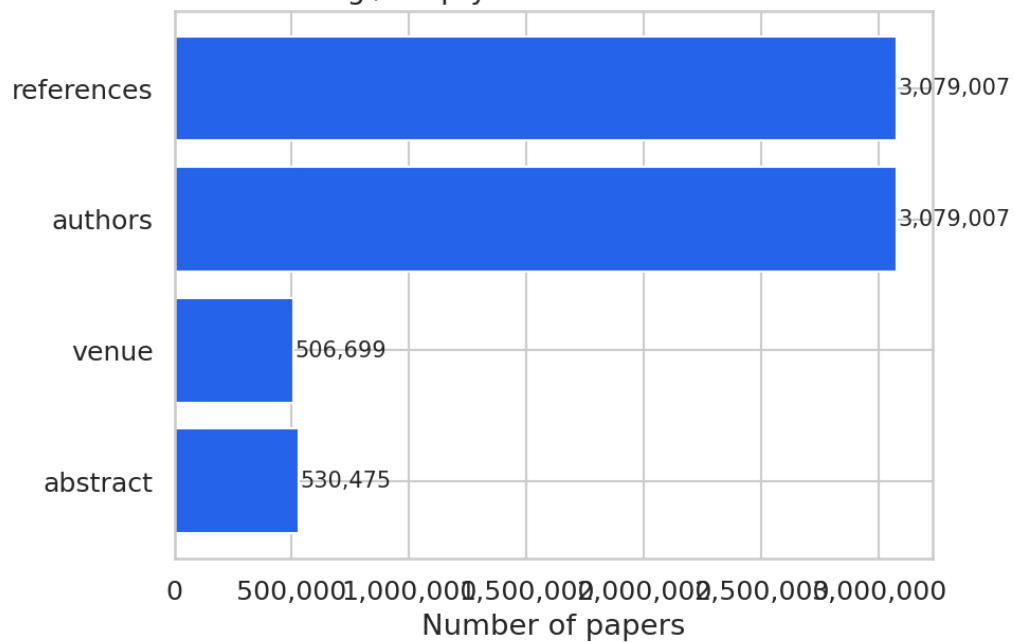


Median Citation Count by Publication Year
(older papers have had more time to accumulate citations)

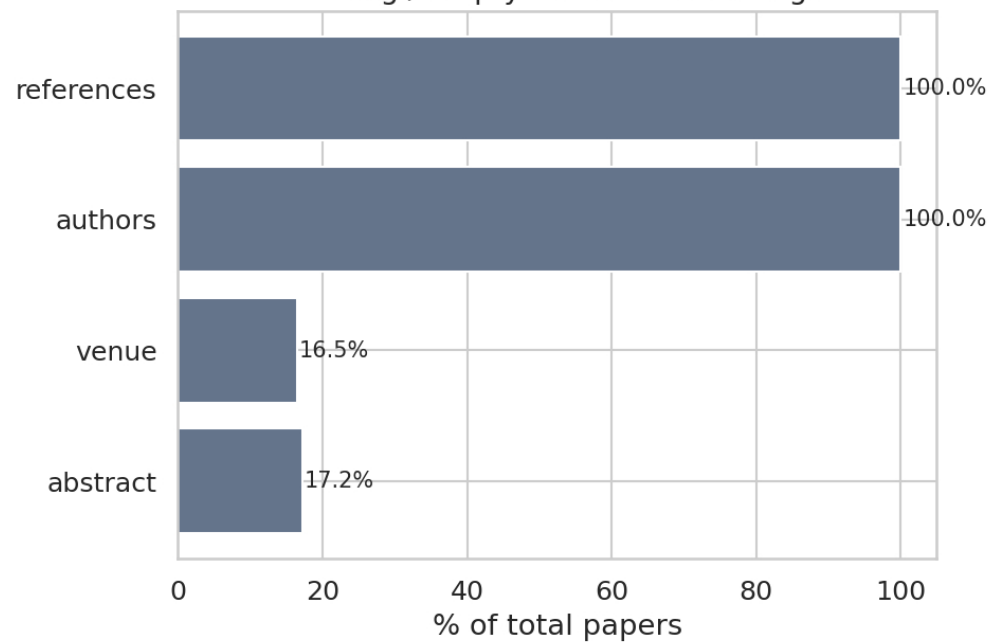


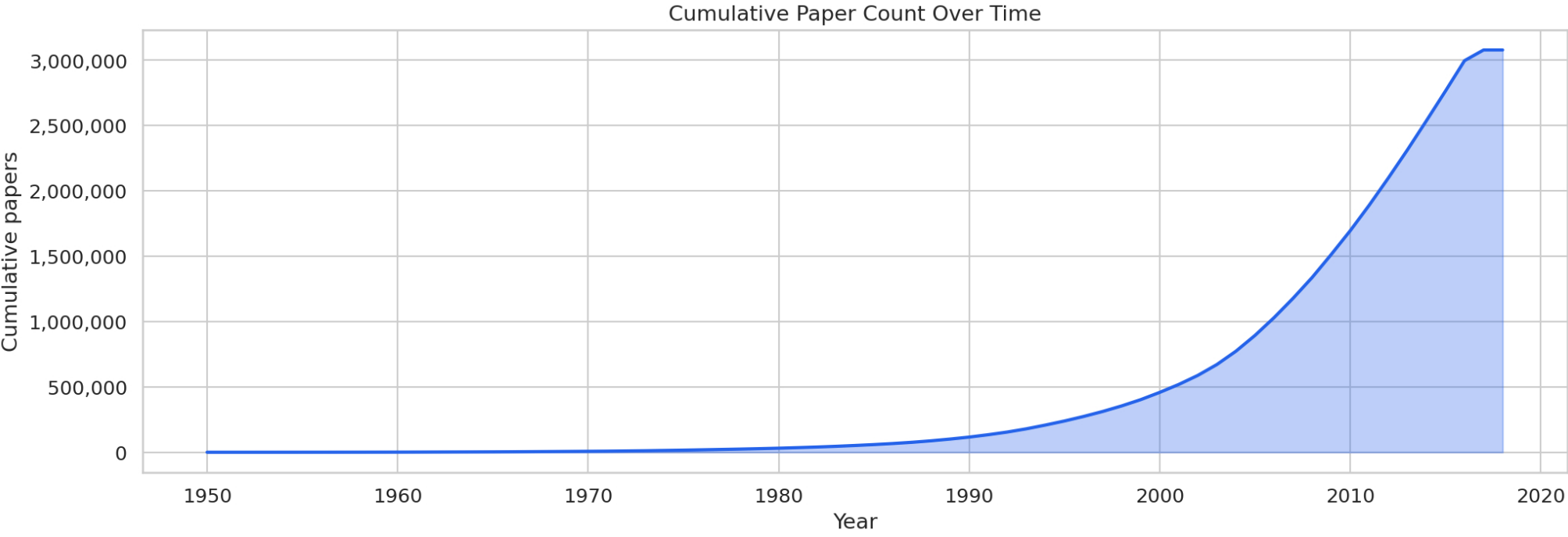
Data Completeness Overview

Missing / Empty Fields — Absolute Count

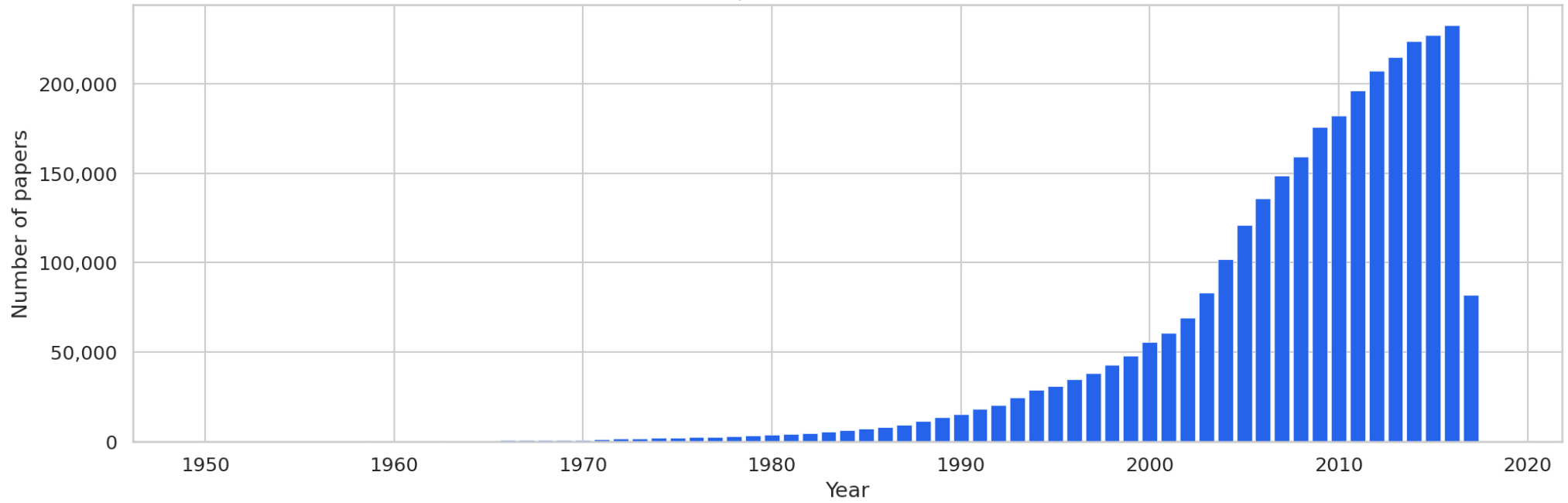


Missing / Empty Fields — Percentage

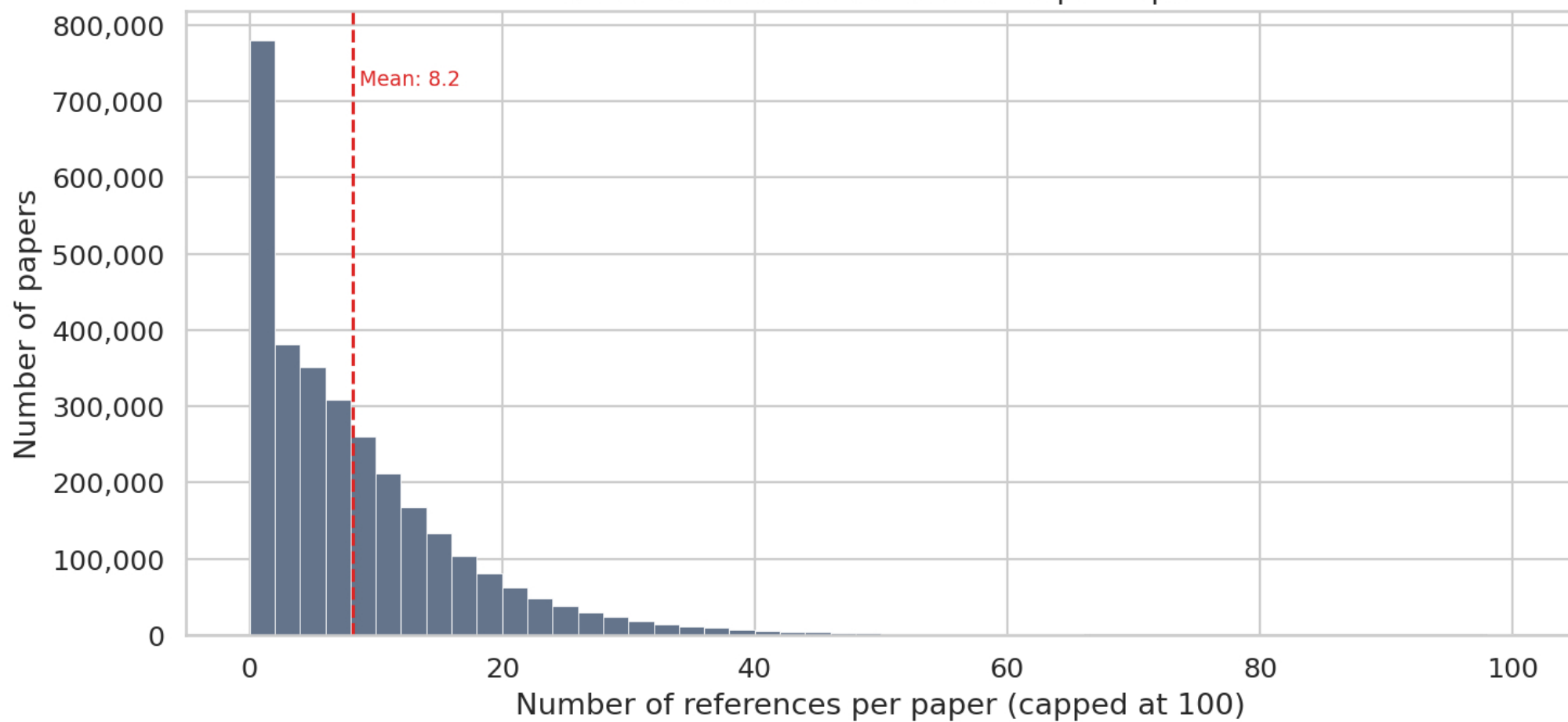




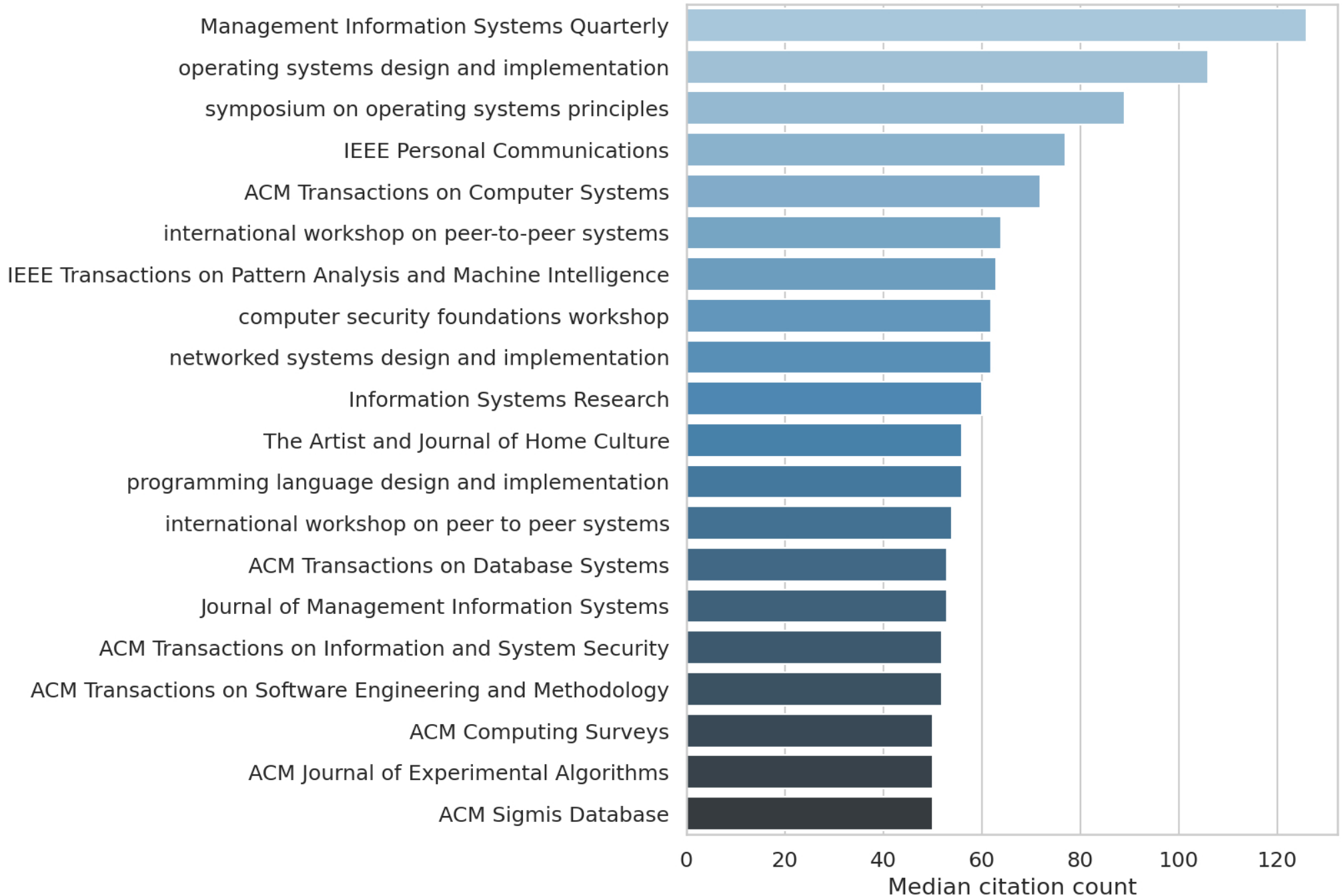
Papers Published Per Year



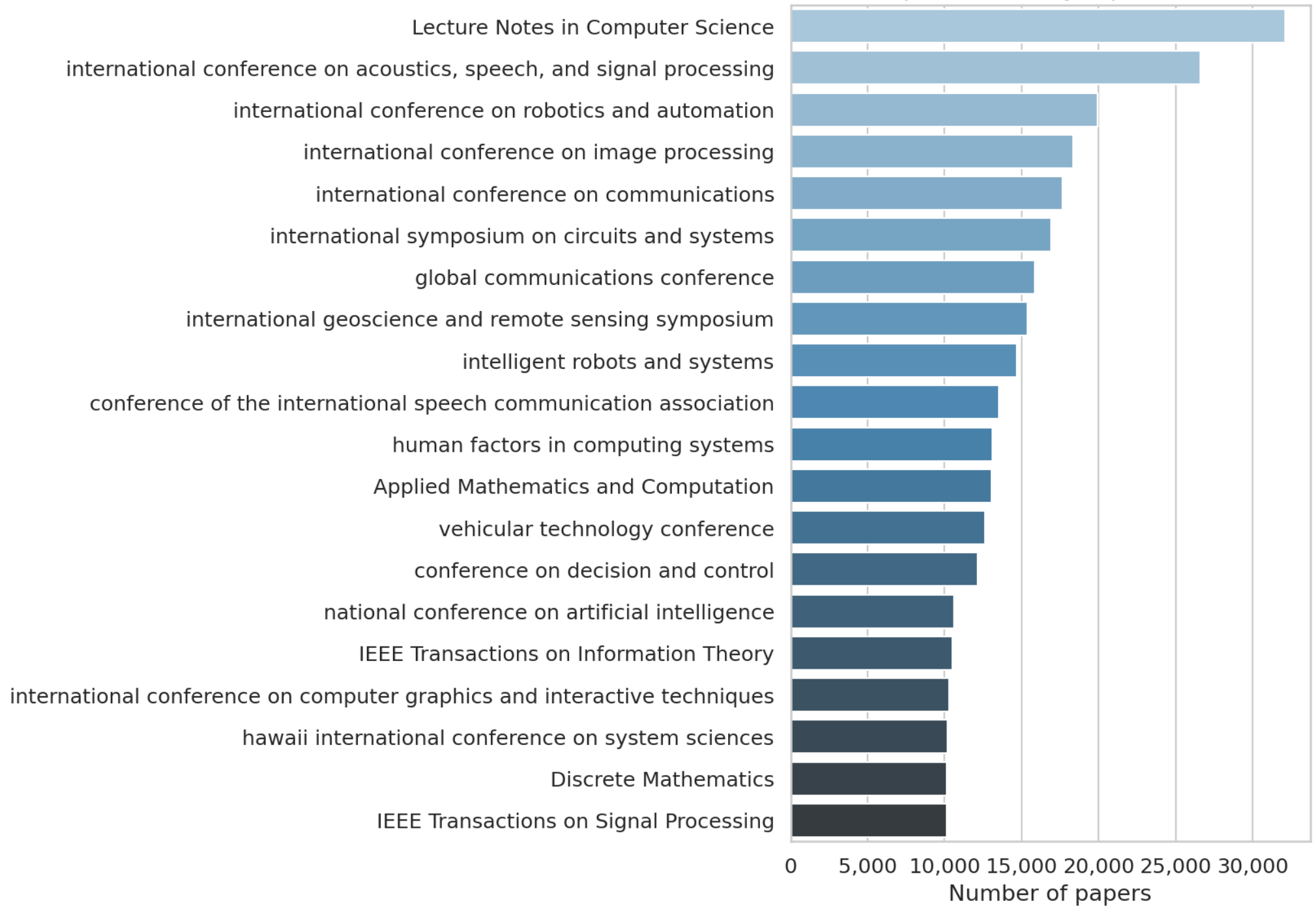
Distribution of Reference Counts per Paper



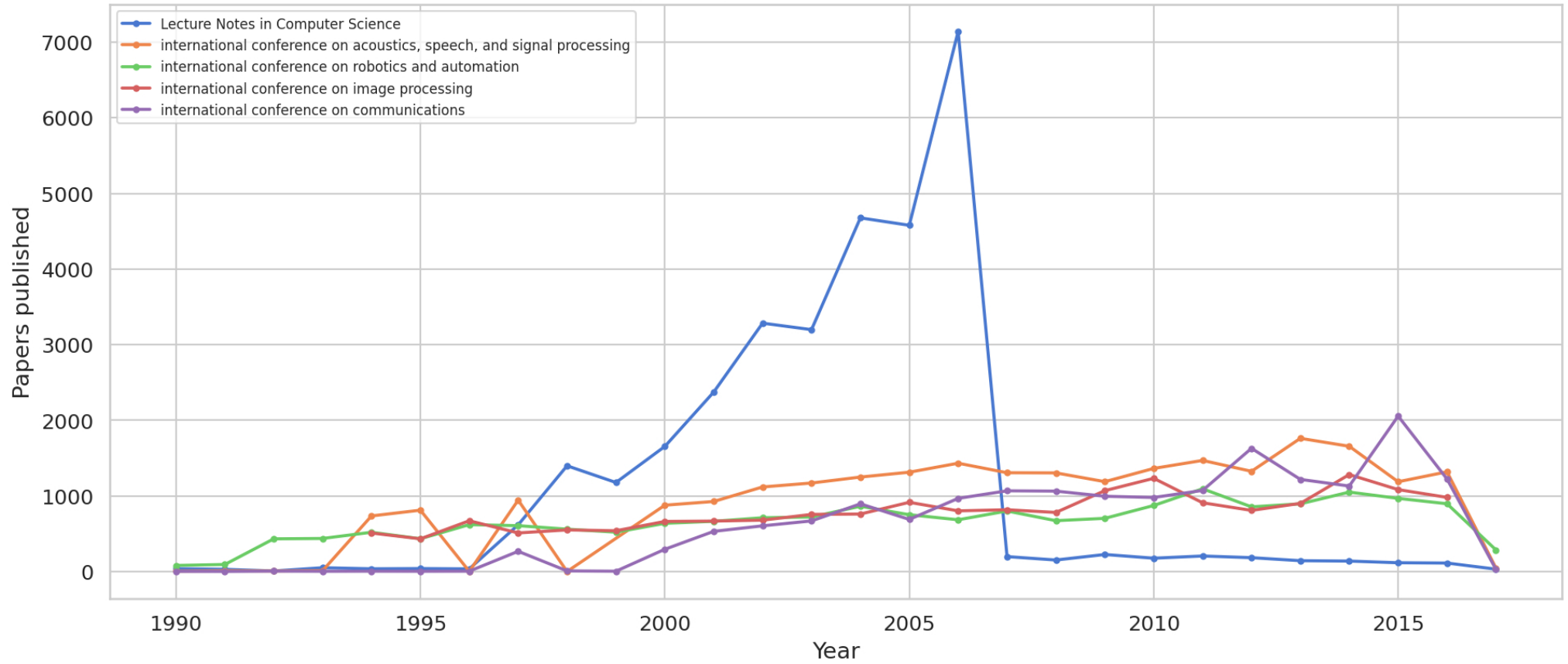
Top 20 Venues by Median Citation Count
(venues with ≥ 50 papers)

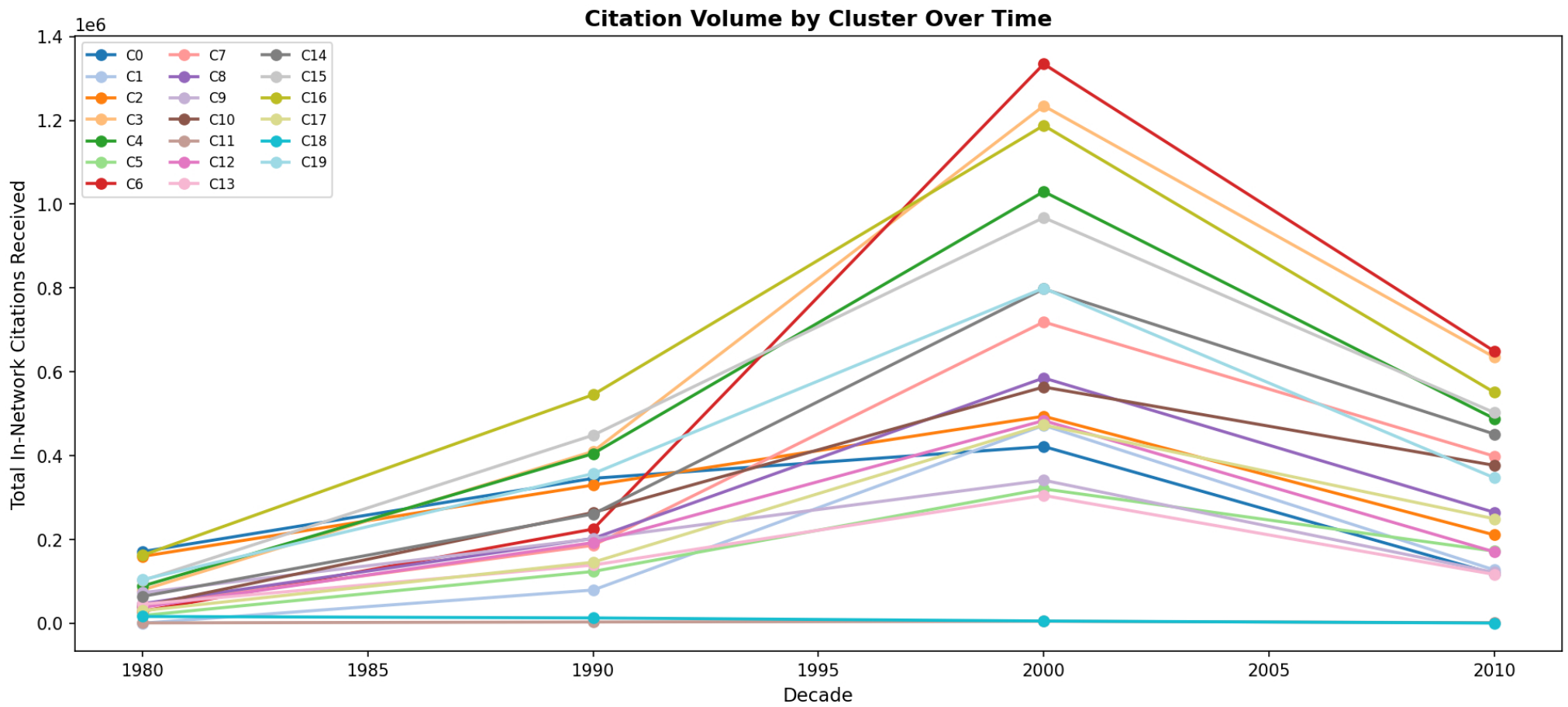


Top 20 Venues by Paper Count

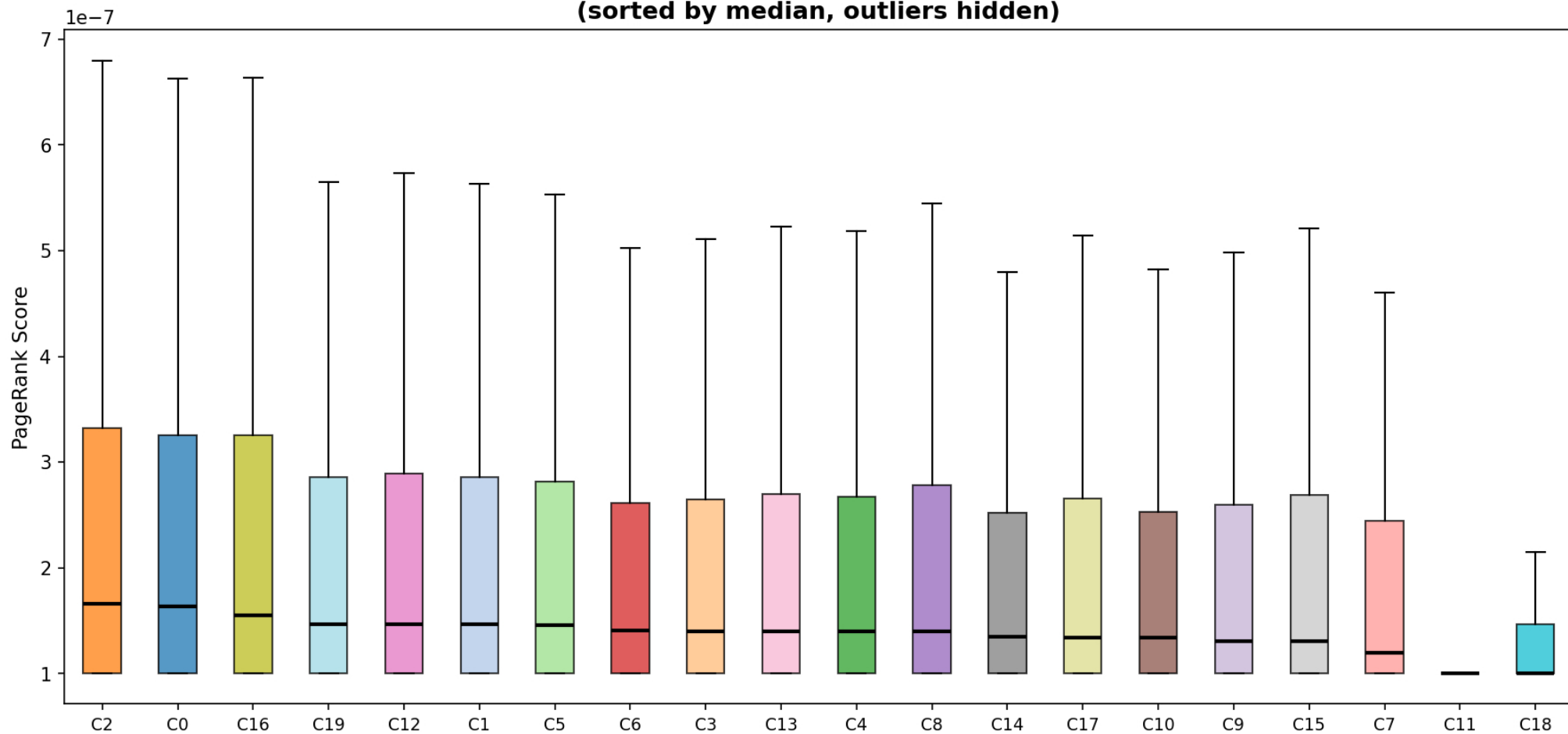


Paper Output Over Time — Top 5 Venues

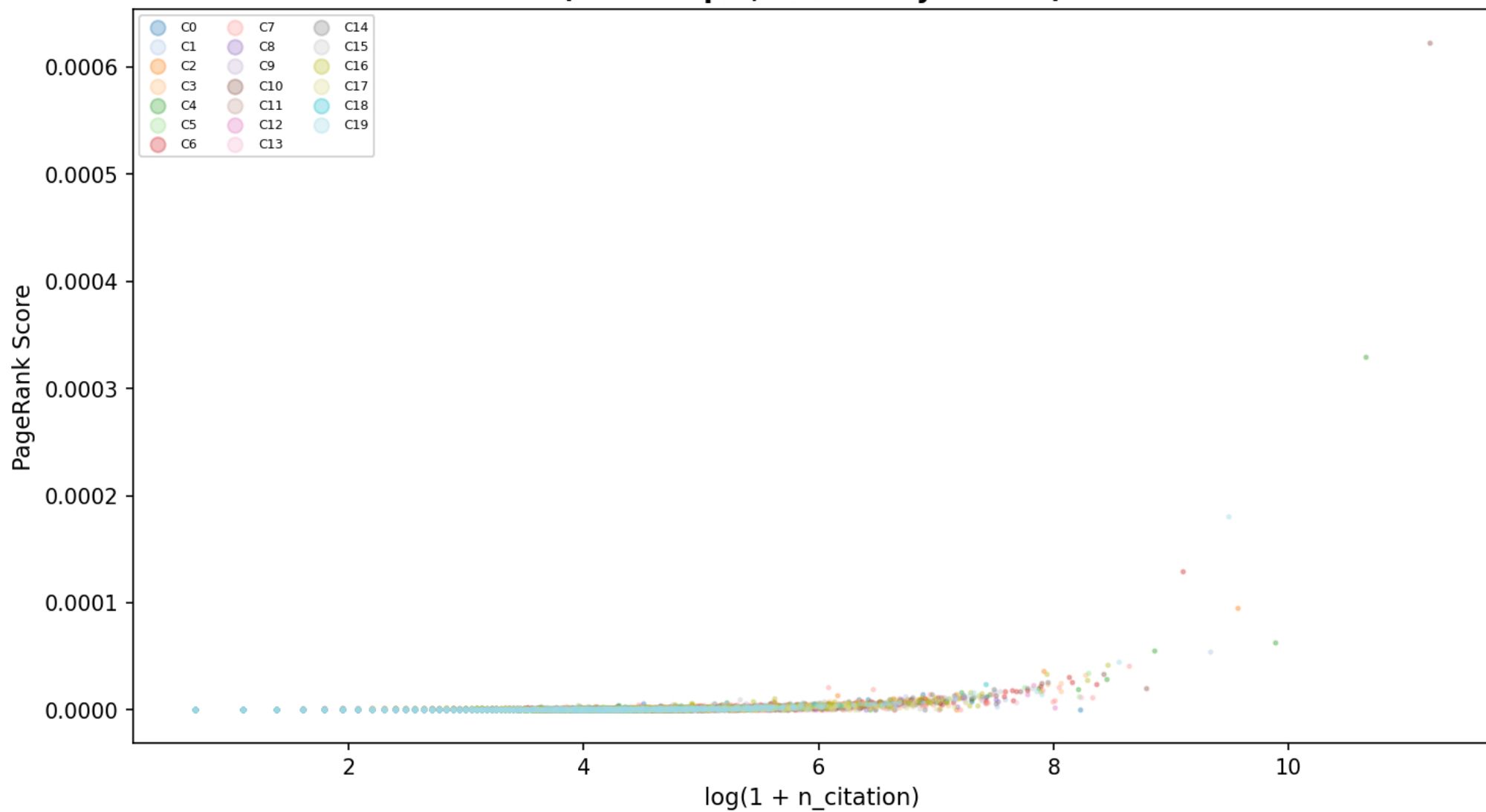




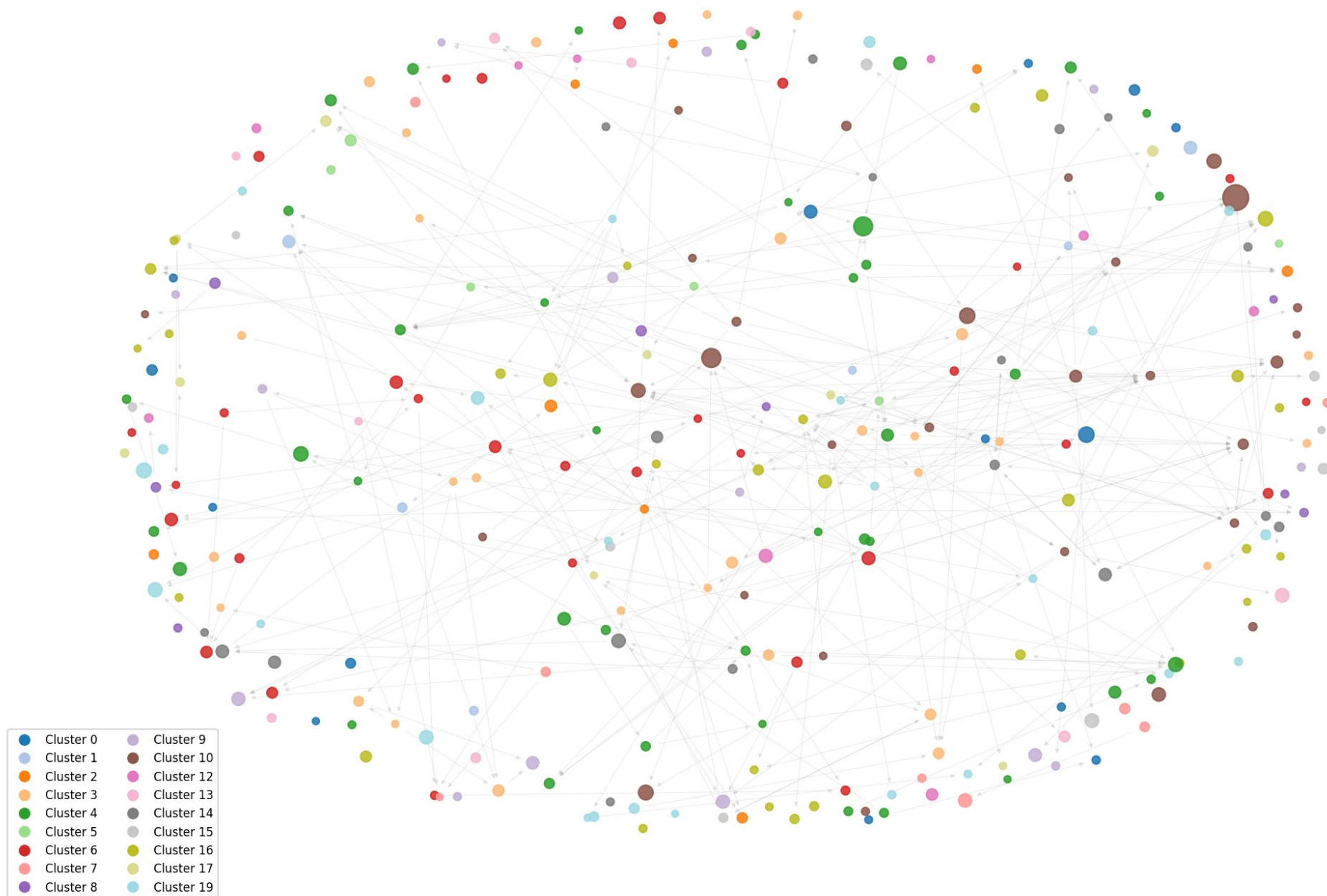
PageRank Distribution per Cluster
(sorted by median, outliers hidden)



**PageRank vs Citation Count
(80k sample, colored by cluster)**



Citation Subgraph — Top 300 Papers by PageRank
(node size $\propto$ PageRank, color = cluster)



Top 20 Papers by PageRank (colored by cluster)

